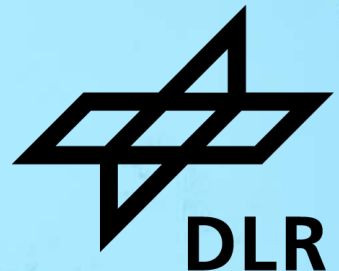


INTRODUCTION TO DEEP LEARNING

PART V – LATEST APPLICATIONS, CODE AND KNOWLEDGE SOURCES

Auliya Fitri, Sheetal Sahoo, Sreerag Naveenachandran

**Machine Learning Group
Institute of Data Science**



Schedule



Date	Time	Activity
28.05.2026 Day 1	09:00 - 10:00	Introduction and basics
	10:00 - 10:30	Hands-on I
	10:30 - 10:45	Coffee break
	10:45 - 11:15	Hands-on II
	11:15 - 12:00	Hands-on III (Group Work)
	12:00 - 13:00	Lunch break
	13:00 - 14:00	Advanced concept and Convolutional Neural Network
	14:00 - 14:45	Hands-on IV
	14:45 - 15:15	Recap Day 1
29.05.2026 Day 2	09:00 - 10:00	Deep Generative Models
	10:00 - 10:45	Hands-on III
	10:45 - 11:15	Latest applications, code and knowledge sources
	11:15 - 11:30	Closing



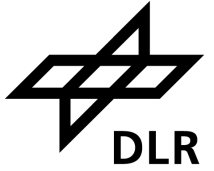
New frontiers

- Transformer
- Diffusion models

Diffusion model

Generative modeling

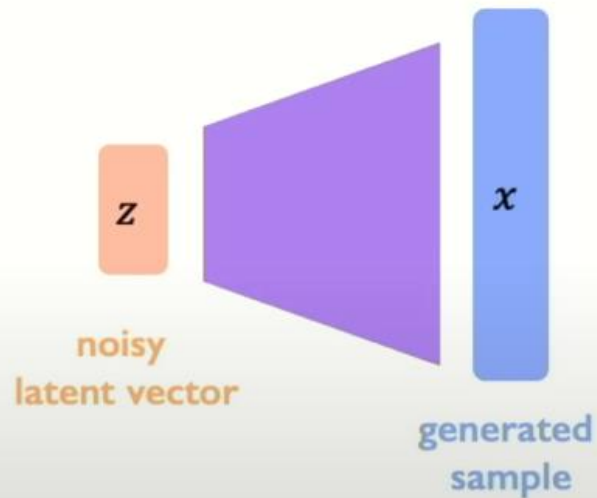
Diffusion models



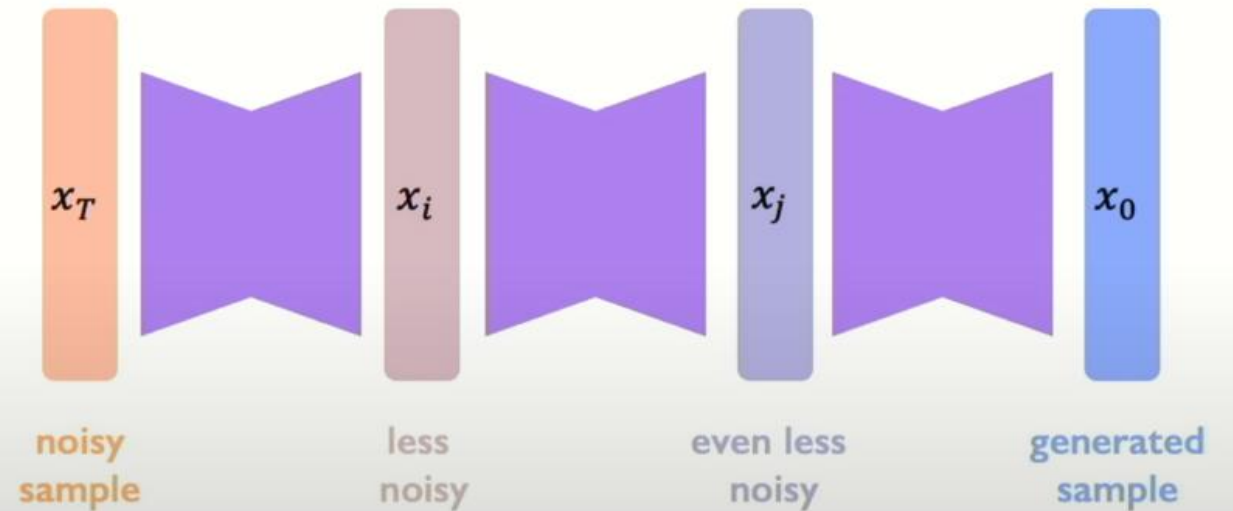
Generative modeling

Diffusion models

VAEs/GANs: Generating samples in one-shot directly from low-dimensional latent variables



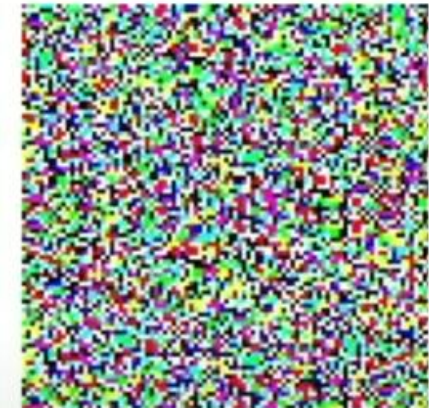
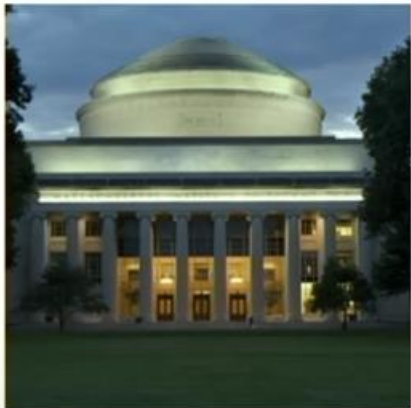
Diffusion: Generating samples iteratively by repeatedly refining and removing noise



Diffusion models

Diffusion process

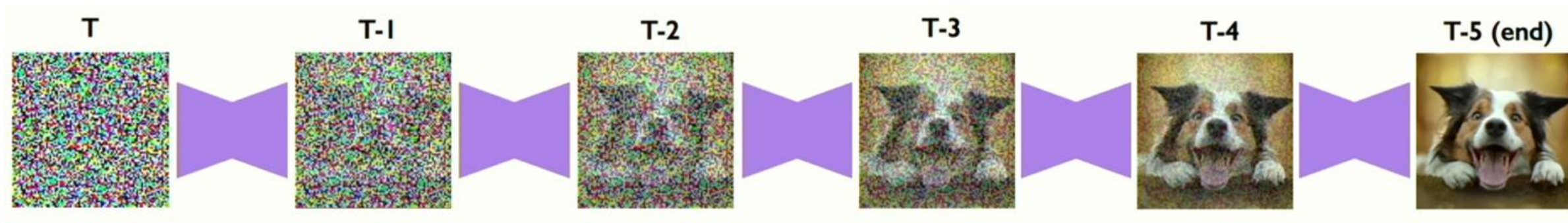
Forward noising
(data-to-noise)



Reverse denoising
(noise-to-data)

Diffusion models

Generating new samples

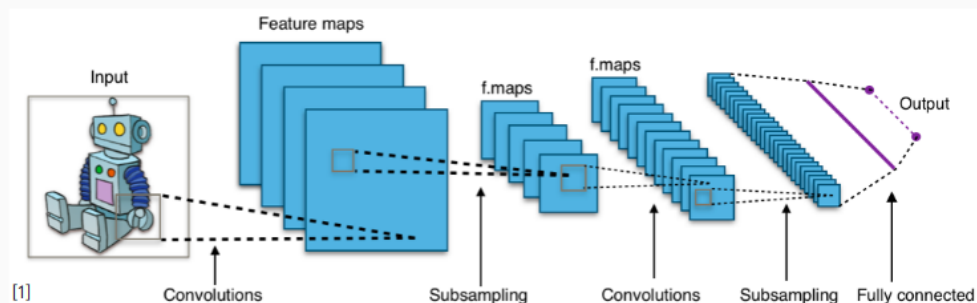


The classic landscape:

One architecture per "community"

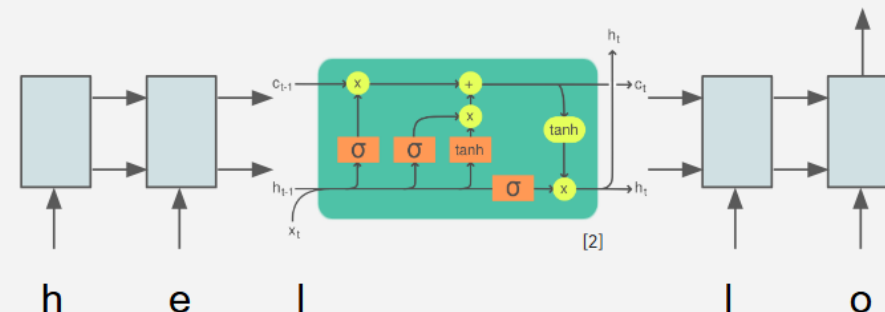
Computer Vision

Convolutional NNs (+ResNets)



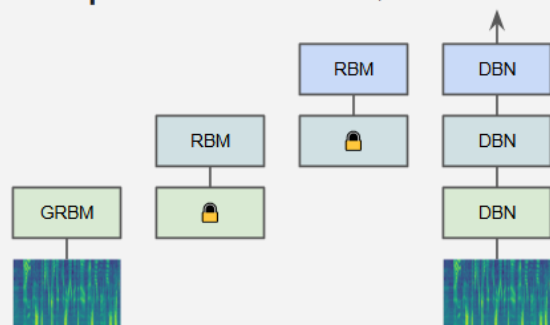
Natural Lang. Proc.

Recurrent NNs (+LSTMs)



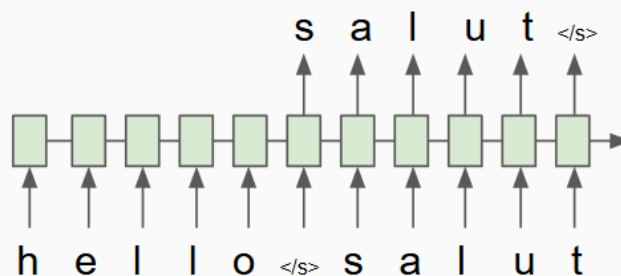
Speech

Deep Belief Nets (+non-DL)



Translation

Seq2Seq



RL

DQN/BC/GAIL

Algorithm 1 Generative adversarial imitation learning

- 1: **Input:** Expert trajectories $\tau_E \sim \pi_E$, initial policy and discriminator parameters θ_0, w_0
- 2: **for** $i = 0, 1, 2, \dots$ **do**
- 3: Sample trajectories $\tau_i \sim \pi_{\theta_i}$
- 4: Update the discriminator parameters from w_i to w_{i+1} with the gradient

$$\hat{\mathbb{E}}_{\tau_i} [\nabla_w \log(D_w(s, a))] + \hat{\mathbb{E}}_{\tau_i} [\nabla_w \log(1 - D_w(s, a))] \quad (17)$$

- 5: Take a policy step from θ_i to θ_{i+1} , using the TRPO rule with cost function $\log(D_{w_{i+1}}(s, a))$. Specifically, take a KL-constrained natural gradient step with

$$\hat{\mathbb{E}}_{\tau_i} [\nabla_{\theta} \log \pi_{\theta}(a|s) Q(s, a)] - \lambda \nabla_{\theta} H(\pi_{\theta}), \quad (18)$$

where $Q(\bar{s}, \bar{a}) = \hat{\mathbb{E}}_{\tau_i} [\log(D_{w_{i+1}}(s, a)) | s_0 = \bar{s}, a_0 = \bar{a}]$

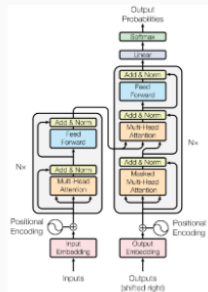
6: **end for**

[1] CNN image CC-BY-SA by Aphex34 for Wikipedia https://commons.wikimedia.org/wiki/File:Typical_cnn.png

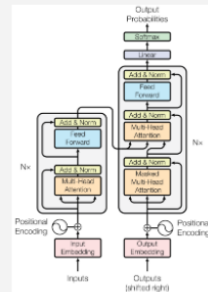
[2] RNN image CC-BY-SA by GChе for Wikipedia https://commons.wikimedia.org/wiki/File:The_LSTM_Cell.svg

The Transformer's takeover: One community at a time

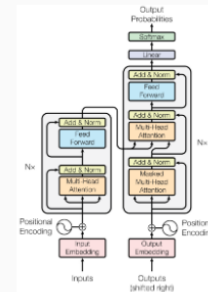
Computer Vision



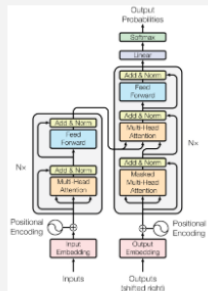
Natural Lang. Proc.



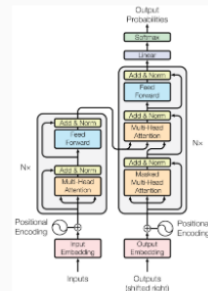
Reinf. Learning



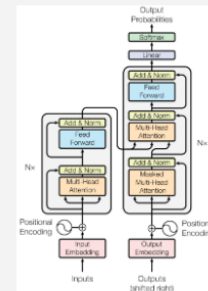
Speech



Translation



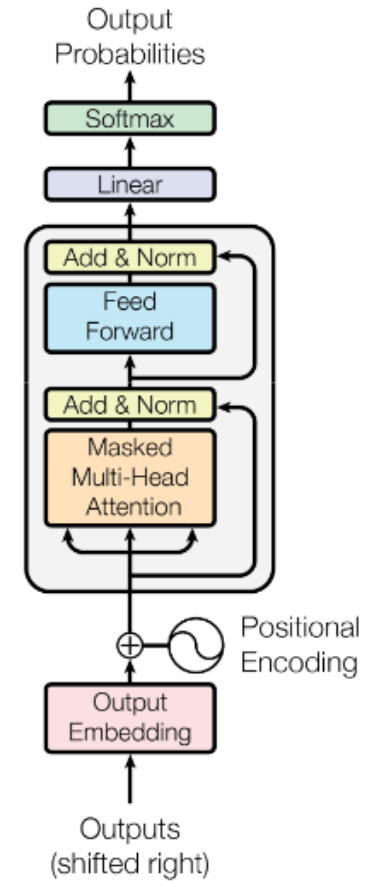
Graphs/Science



Transformer image source: "Attention Is All You Need" paper

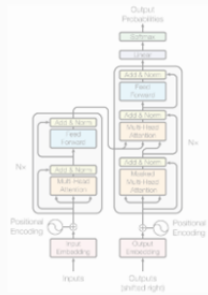
Decoder-only GPT

[sat_]

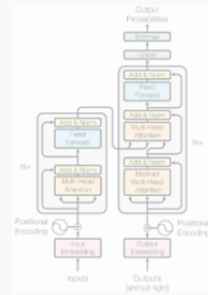


[START] [The_] [cat_]

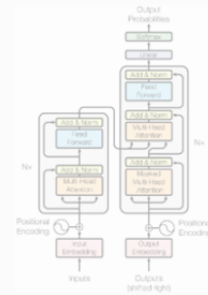
Computer Vision



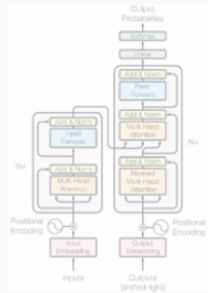
Natural Lang. Proc.



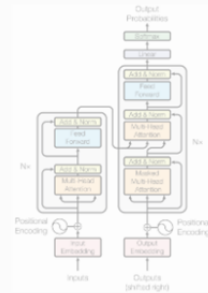
Reinf. Learning



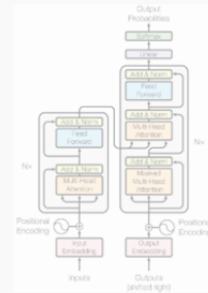
Speech



Translation



Graphs/Science



Transformer image source: "Attention Is All You Need" paper

An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

2020, A Dosovitskiy, L Beyer, A Kolesnikov, D Weissenborn, X Zhai, T Unterthiner, M Dehghani, M Minderer, G Heigold, S Gelly, J Uszkoreit, N Houlsby

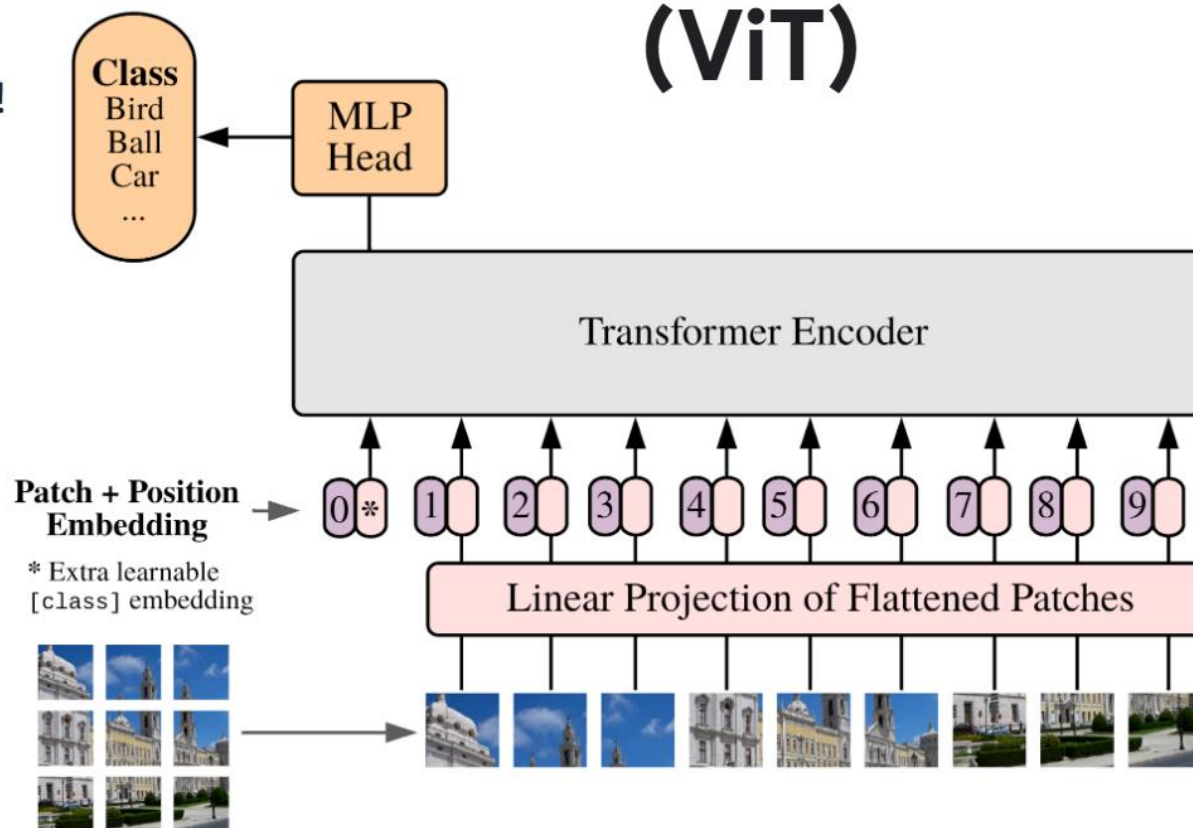
Many prior works attempted to introduce self-attention at the pixel level.

For 224px^2 , that's 50k sequence length, too much!

Thus, most works restrict attention to local pixel neighborhoods, or as high-level mechanism on top of detections.

The **key breakthrough** in using the full Transformer architecture, standalone, was to **"tokenize" the image** by **cutting it into patches** of 16px^2 , and treating each patch as a token, e.g. embedding it into input space.

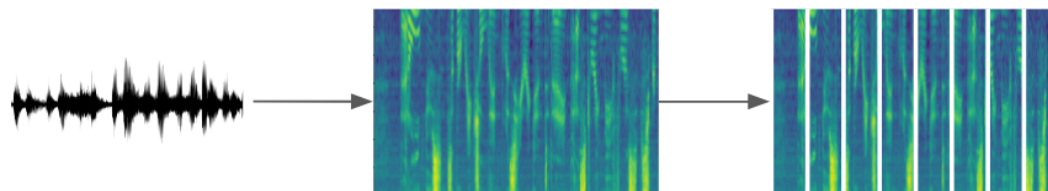
Vision Transformer (ViT)



Whisper: Robust Speech Recognition via Large-Scale Weak Supervision

2022, A Radford, J W Kim, T Xu, G Brockman, C McLeavey, I Sutskever

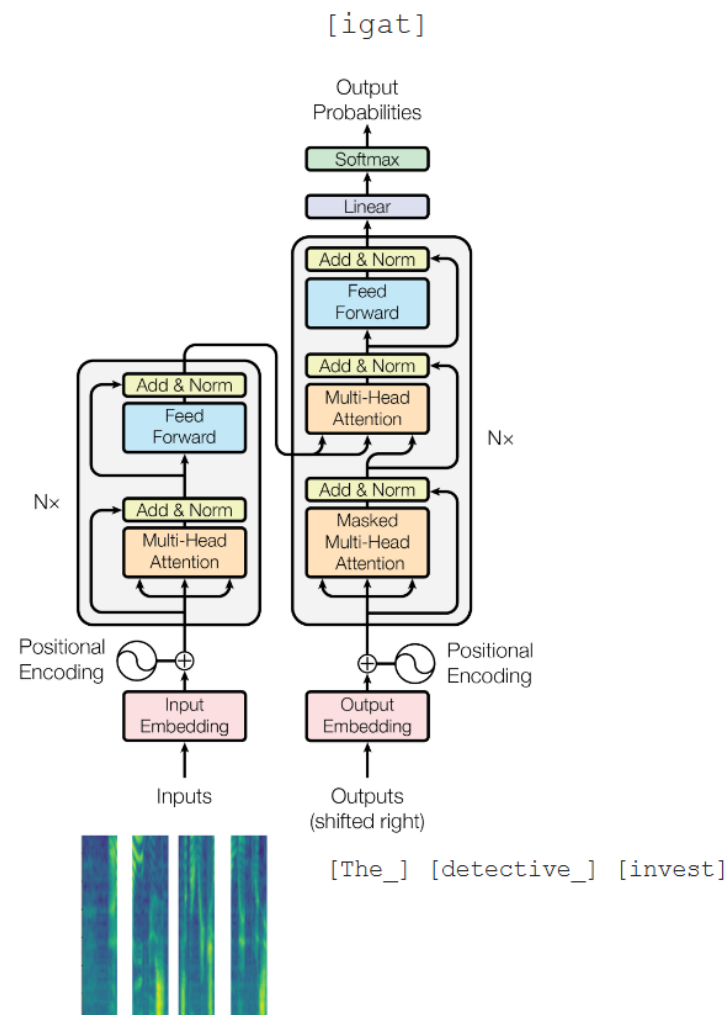
Largely the same story as in computer vision.
But with spectrograms instead of images.



Conformer adds a third type of block using convolutions, and slightly reorder blocks, but overall very transformer-like.

Exists as encoder-decoder variant, or as encoder-only variant with CTC loss.

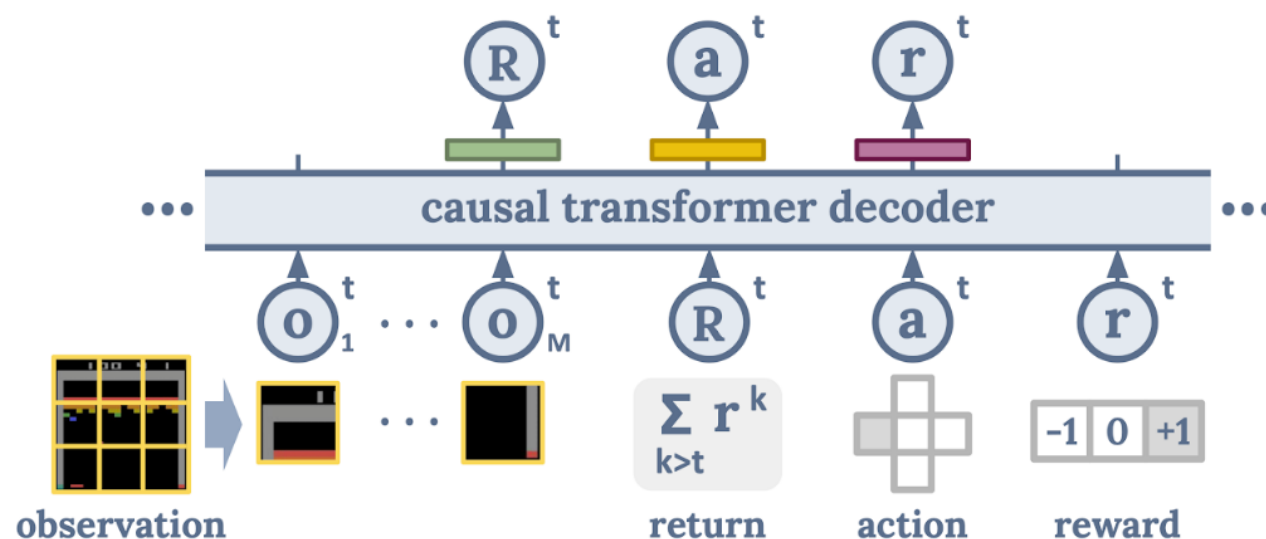
UPDATE Sept 2022: Plain transformer model in “Whisper” by
A Radford, J W Kim, T Xu, G Brockman, C McLeavey, I Sutskever



Decision Transformer: Reinforcement Learning via Sequence Modeling

2021, L Chen, K Lu, A Rajeswaran, K Lee, A Grover, M Laskin, P Abbeel, A Srinivas, I Mordatch

Cast the (supervised/offline) RL problem into a sequence ("language") modeling task:



Can generate/decode sequences of actions with desired return (eg skill)

The trick is prompting: "The following is a trajectory of an expert player: [obs] ..."

Anything

ca 2023 and onwards

you can Generate!

For outputs: learn discrete code (VQ-VAE, FSQ) that represents the output, append to vocab.

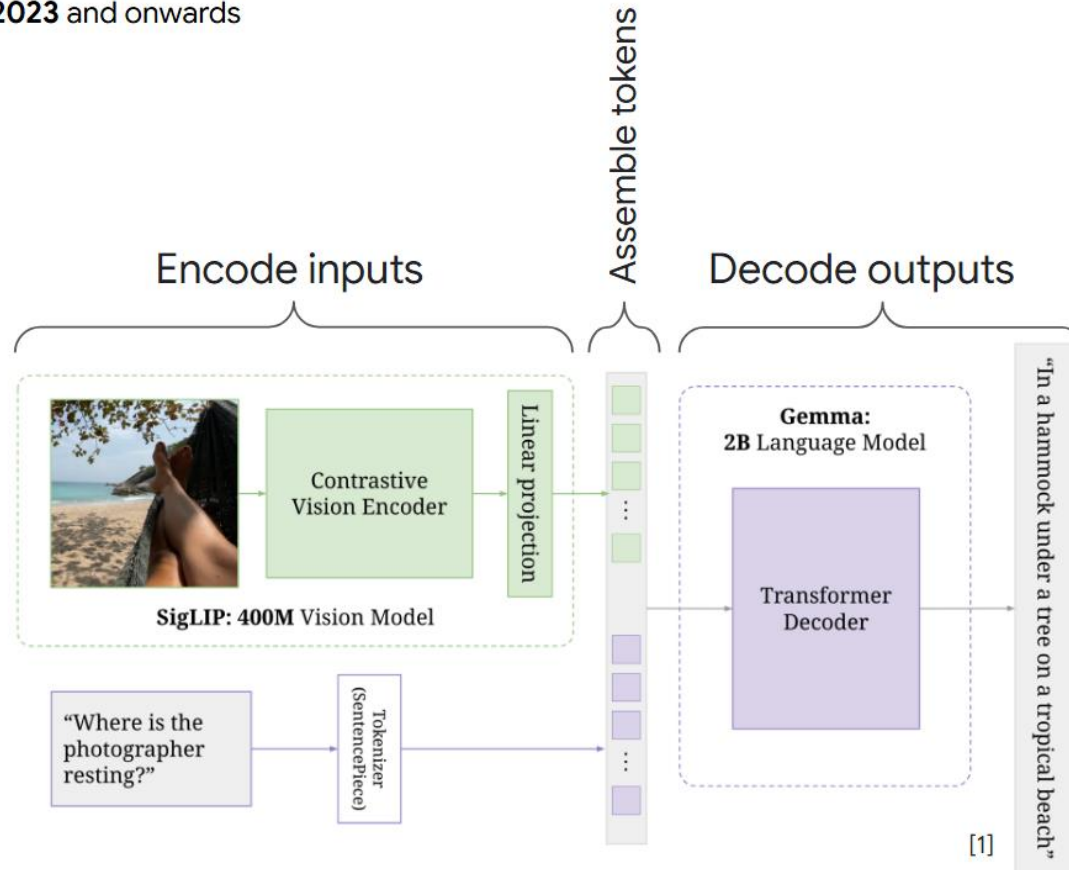
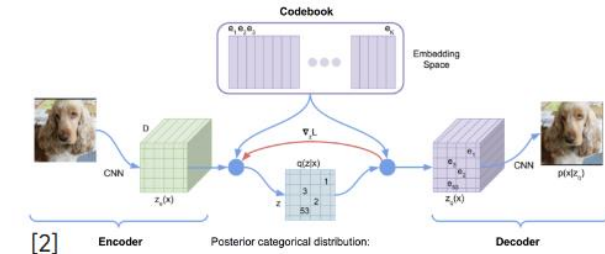


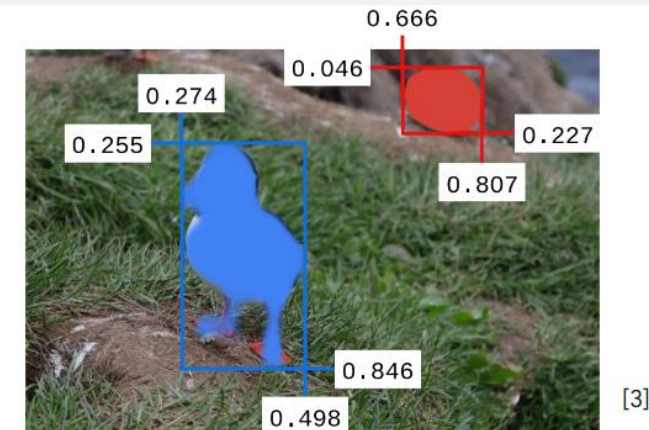
Figure 1 | PaliGemma's architecture: a SigLIP image encoder feeds into a Gemma decoder LM.

- [1] PaliGemma by L. Beyer, A. Steiner, A. S. Pinto, A. Kolesnikov, X. Wang, ..., X. Zhai
- [2] VQ-VAE by A. van den Oord, O. Vinyals, K. Kavukcuoglu
- [3] PaliGemma 2 by Andreas Steiner, A. S. Pinto, M. Tschannen, ..., L. Beyer, X. Zhai



Input:

segment puffin in the back ; puffin in front



Output:

```
<loc0255><loc0274><loc0846><loc0498>
<seg024>[...]<seg018> puffin in front ;
<loc0046><loc0666><loc0227><loc0807>
<seg106>[...]<seg055> puffin in the back
```


Diffusion models

Text to image generation

“a painting of a fox sitting in a field at sunrise in the style of Claude Monet”



“an ibis in the wild, painted in the style of John Audubon”



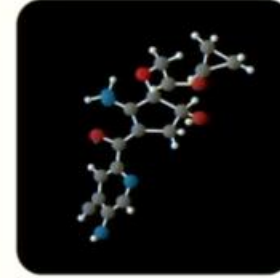
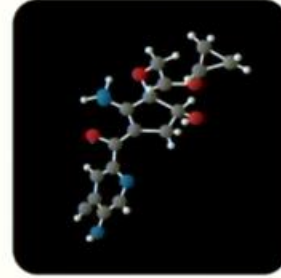
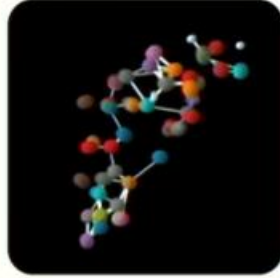
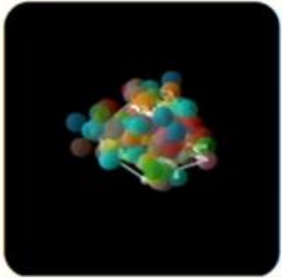
“close-up of a snow leopard in the snow hunting, rack focus, nature photography”



Diffusion models

Beyond images: Molecular Design

Chemistry: Generating Molecules in 3D

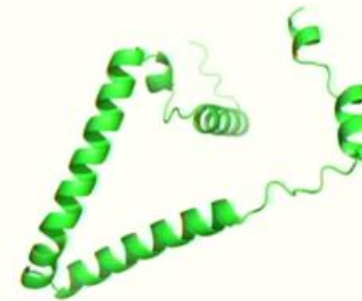
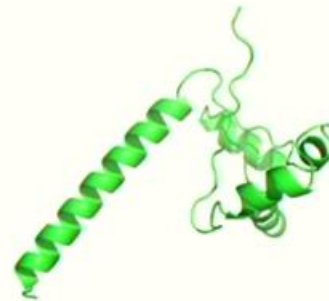
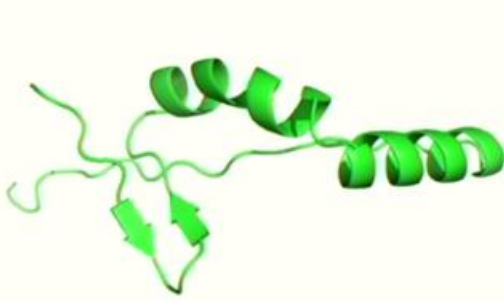
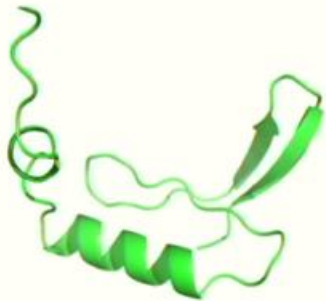


Noise

Molecule

Hoogeboom+ ICML 2022, Jing+ NeurIPS 2022.

Biology: Generating Novel Proteins



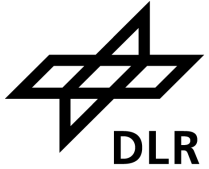
Wu+ arXiv 2022, Anand+ arXiv 2022, Trippe+ arXiv 2022, Jumper+ arXiv 2022, and more ...

Knowledge sources

Code & knowledge sources

Code: Github

- Many free implementations of popular and new methods
- Many conferences now require public code for reproducibility
- Usually a good first step for testing and adapting a new method



The screenshot shows the GitHub repository page for 'ultralytics/yolov5'. The browser address bar shows the URL 'https://github.com/ultralytics/yolov5'. The repository has tabs for 'README', 'Code of conduct', 'Contributing', 'AGPL-3.0 license', and 'Security'. The 'Documentation' section is expanded, showing instructions for installation and inference. The installation instructions include cloning the repository, navigating to the directory, and installing dependencies. The inference instructions show a Python script for loading a model and performing inference on an image.

```
# Clone the YOLOv5 repository
git clone https://github.com/ultralytics/yolov5

# Navigate to the cloned directory
cd yolov5

# Install required packages
pip install -r requirements.txt
```

```
import torch

# Load a YOLOv5 model (options: yolov5n, yolov5s, yolov5m, yolov5l, yolov5x) # Default is yolov5s
model = torch.hub.load("ultralytics/yolov5", "yolov5s")

# Define the input image source (URL, local file, PIL image, etc.) # Example
img = "https://ultralytics.com/images/zidane.jpg"

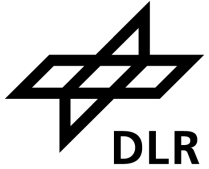
# Perform inference (handles batching, resizing, normalization)
results = model(img)

# Process the results (options: .print(), .show(), .save(), etc.)
results.print() # Print results to console
results.show() # Display results in a window
results.save() # Save results to runs/detect/exp
```


Code & knowledge sources

Knowledge: arXiv

- Now often the first point of publication for DL papers (sometimes the only one)
- Not peer-reviewed!
- Helpful tool:
- <http://www.arxiv-sanity.com/>



A screenshot of a web browser displaying an arXiv paper page. The browser's address bar shows the URL 'https://arxiv.org/abs/2407.20892v1'. The page header includes the Cornell University logo and name. The paper title is 'What is YOLOv5: A deep look into the internal features of the popular object detector' by Rahima Khanam and Muhammad Hussain, submitted on 30 Jul 2024. The page offers two main options: 'View PDF' and 'HTML (experimental)'. A summary paragraph describes the study's focus on the YOLOv5 model's architecture and performance. Below this, it lists subjects as 'Computer Vision and Pattern Recognition (cs.CV)' and provides citation information, including the arXiv ID and a DOI link. A 'Submission history' section shows the paper was submitted by Rahima Khanam on Tue, 30 Jul 2024 at 15:09:45 UTC. At the bottom, there are buttons for 'View PDF', 'HTML (experimental)', and 'TeX Source', along with a Creative Commons BY license icon and navigation links for previous/next papers and browsing by date.

Code & knowledge sources

Knowledge: Blogs



The Tensor



The Loss Function



The Autograd



The nn Module

How PyTorch lets you build and experiment with a neural net

We show, an example of building a classifier neural network in PyTorch and highlight how easy it is to experiment with advanced ideas.

 Tirthajyoti Sarkar in Towards Data Science
Nov 27 · 10 min read ★

LATEST

Sound classification using Images
Learn how to classify audio files using spectrograms

Dipam Vasani in Towards Data Science
May 14 · 5 min read ★

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Teaching the learners.

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<https://medium.com/topic/artificial-intelligence>
<https://medium.com/topic/machine-learning>

Real-time Object Detection with YOLO, YOLOv2 and now YOLOv3



Jonathan Hui

Follow

Mar 18, 2018 · 18 min read



You only look once (YOLO) is an object detection system targeted for real-time processing. We will introduce YOLO, YOLOv2 and YOLO9000 in this article. For those only interested in YOLOv3, please forward to the [bottom of the article](#). Here is the accuracy and speed comparison provided by the YOLO web site.

Model	Train	Test	mAP	FLOPS	FPS
SSD300	COCO trainval	test-dev	41.2	-	46
SSD500	COCO trainval	test-dev	46.5	-	19
YOLOv2 608x608	COCO trainval	test-dev	48.1	62.94 Bn	40
Tiny YOLO	COCO trainval	-	-	7.07 Bn	200
SSD321	COCO trainval	test-dev	45.4	-	16
DSSD321	COCO trainval	test-dev	46.1	-	12
R-FCN	COCO trainval	test-dev	51.9	-	12
SSD513	COCO trainval	test-dev	50.4	-	8
DSSD513	COCO trainval	test-dev	53.3	-	6
FPN FRCN	COCO trainval	test-dev	59.1	-	6
RetinaNet 50 500	COCO trainval	test-dev	59.9	-	14

Code & knowledge sources

Code: Conferences (selection)

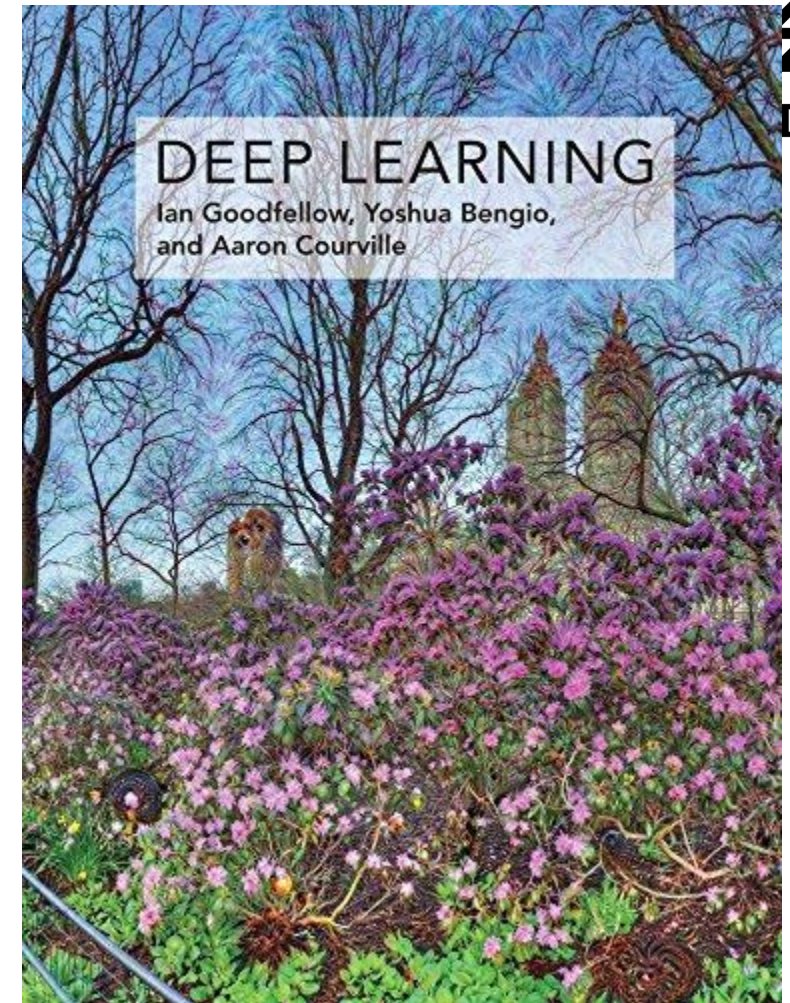
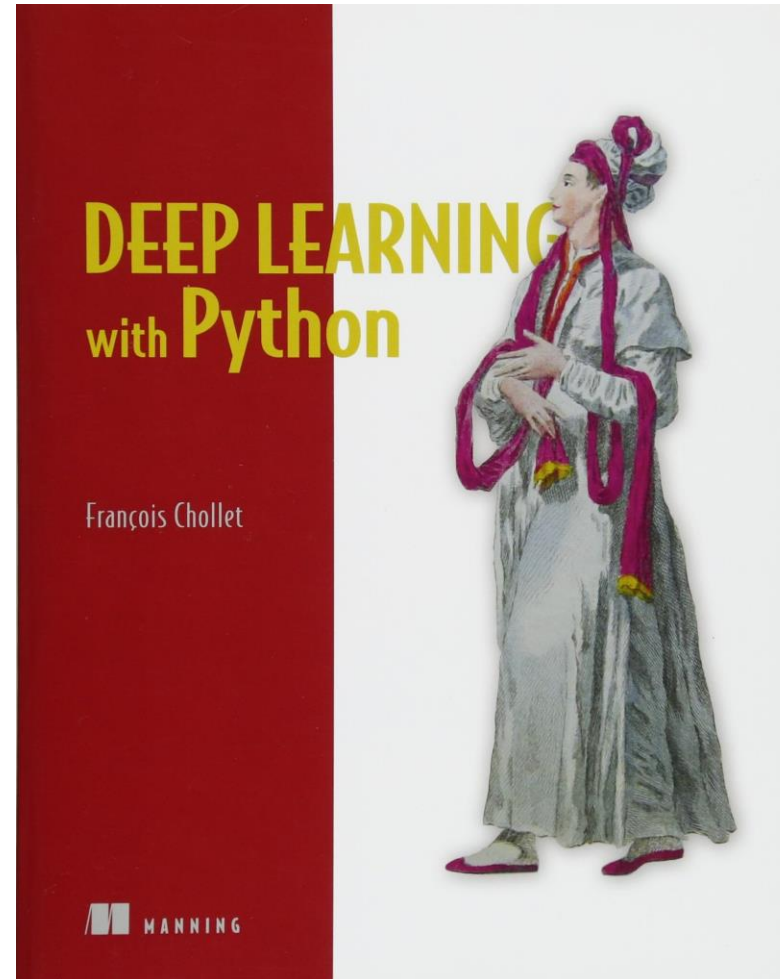
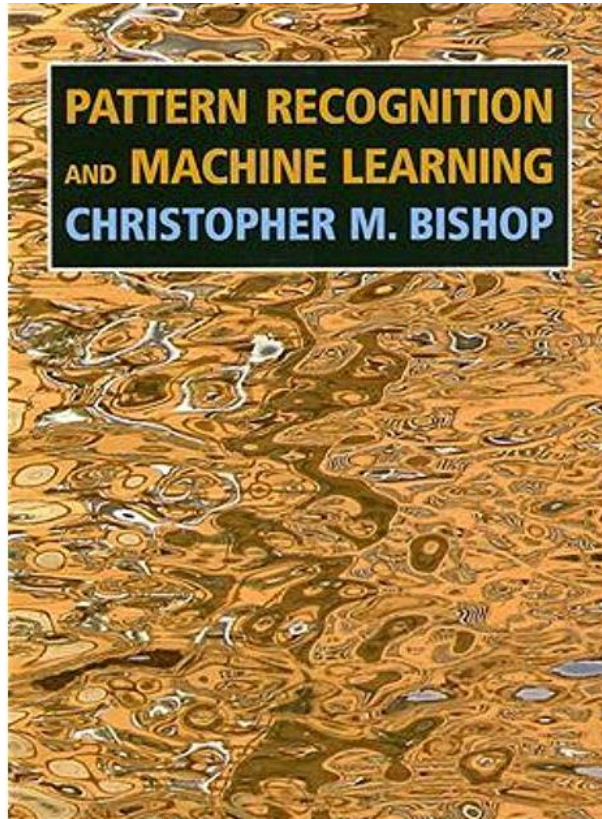


...and of course
many domain-
specific conferences



Code & knowledge sources

Knowledge: Books



<http://www.deeplearningbook.org/>

Code & knowledge sources

Knowledge: Online Courses



Browse > Data Science > Machine Learning

Offered By

Machine Learning

★★★★★ 4.9 120,193 ratings • 29,510 reviews

Enroll for Free
Started Nov 25

Financial aid available

2,655,317 already enrolled!

About Syllabus Reviews Instructors Enrollment Options FAQ

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Offered By

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deeplearning.ai

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7,781,080 recent views

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Machine Learning

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387,877 recent views

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Deep Learning



100% online courses

Start instantly and learn at your own schedule.



Flexible Schedule

Set and maintain flexible deadlines.



Intermediate Level



Approx. 3 months to complete

Suggested 11 hours/week

<https://www.coursera.org/>

A.Fitri, S.Sahoo, S.Naveenachandran, DLR-DW, 28.-29.05.2026

Thank you for your participation!



Auliya Fitri, Sheetal Sahoo, Sreerag Naveenachandran
Machine Learning Group
Institute of Data Science

Topic: **Introduction to Deep Learning**
Part V – Code and Knowledge Sources

Date: 2025-05-29

Author: Auliya Fitri, Sheetal Sahoo, Sreerag Naveenachandran

Institute: Data Science

Image sources: All images “DLR (CC BY-NC-ND 3.0)” unless otherwise stated